Definitions of adverse events associated with extracorporeal membrane oxygenation in children: results of an international Delphi process from the ECMO-CENTRAL ARC

Peta M A Alexander, Matteo Di Nardo, Alain Combes, Adam M Vogel, Marta Velia Antonini, Nicholas Barrett, Giulia M Benedetti, Amanda Bettencourt, Daniel Brodie, René Gómez-Gutiérrez, Stephen M Gorga, Carol Hodgson, Poonam Malhotra Kapoor, Jennifer Le, Graeme MacLaren, Erika R O'Neil, Marlies Ostermann, Matthew L Paden, Neil Patel, Alvaro Rojas-Peña, Ahmed S Said, Francesca Sperotto, Ariane Willems, Leen Vercaemst, Ajit P Yoganathan, Angela Lorts, Pedro J del Nido, Ryan P Barbaro for the endorsing societies of Extracorporeal Life Support Organization (ELSO), International ECMO Network (ECMONet), Pediatric Acute Lung Injury and Sepsis Investigators (PALISI), Pediatric ECMO subgroup of PALISI and ELSO (PediECMO), European Society of Paediatric and Neonatal Intensive Care (ESPNIC), Australian and New Zealand Intensive Care Society Paediatric Study Group (ANZICS PSG), Intensive Care Society (ICS), Pediatric Cardiac Intensive Care Society (PCICS), Advanced Cardiac Therapies Improving Outcomes Network (ACTION), Children's Hospitals Neonatal Consortium (CHNC), American Pediatric Surgical Association (APSA), Society of Thoracic Surgeons (STS), Society of Critical Care Medicine (SCCM)

Extracorporeal membrane oxygenation (ECMO) is a high-risk and low-volume life support with increasing clinical study. However, heterogenous outcome definitions impede data assimilation into evidence to guide practice. The Academic Research Consortium (ARC), an international collaborative forum committed to facilitating the creation of stakeholder-driven consensus nomenclature and outcomes for clinical trials of medical devices, supported the ECMO Core Elements Needed for Trials Regulation And quality of Life (ECMO-CENTRAL) ARC. The ECMO-CENTRAL ARC was assembled to develop definitions of paediatric ECMO adverse events for use in clinical trials and regulatory device evaluation. An initial candidate list of ECMO adverse events derived from the mechanical circulatory support ARC was supplemented with a review of ECMO-relevant adverse event definitions collated from literature published between Jan 1, 1988, and Feb 20, 2023. Distinct teams of international topic experts drafted separate adverse event definitions that were harmonised to existing literature when appropriate. Draft definitions were revised for paediatric ECMO relevance with input from patients, families, and an international expert panel of trialists, clinicians, statisticians, biomedical engineers, device developers, and regulatory agencies. ECMO-CENTRAL ARC was revised and disseminated across research societies and professional organisations. Up to three rounds of internet-based anonymous surveys were planned as a modified Delphi process. The expert panel defined 13 adverse event definitions: neurological, bleeding, device malfunction, acute kidney injury, haemolysis, infection, vascular access-associated injury, non-CNS thrombosis, hepatic dysfunction, right heart failure, left ventricular overload, lactic acidaemia, and hypoxaemia. Definitional structure varied. Among 165 expert panel members, 114 were eligible to vote and 111 voted. Consensus was achieved for all proposed definitions. Agreement ranged from 82% to 95%. ECMO-CENTRAL ARC paired rigorous development with methodical stakeholder involvement and dissemination to define paediatric ECMO adverse events. These definitions will facilitate new research and the assimilation of data across clinical trials and ECMO device evaluation in children.

Lancet Child Adolesc Health
2024; 8: 773–80

Department of Cardiology (P M A Alexander MBBS), **Department of Cardiac Surgery** (Prof P J del Nido MD), **and Department of Pediatrics** (F Sperotto MD), **Boston Children's Hospital, Harvard Medical School, Boston, MA, USA; Paediatric Intensive Care Unit, Bambino Gesù Children's Hospital, Scientific Institute for Research, Hospitalization and Healthcare, Rome, Italy** (Prof M Di Nardo MD); **Service de Médecine Intensive-Réanimation, Institut de Cardiologie, APHP Sorbonne Hôpital Pitié–Salpêtrière, Paris, France** (Prof A Combes MD); **Michael E DeBakey Department of Surgery, Baylor College of Medicine, Houston, TX, USA** (Prof A M Vogel MD); **Department of Surgery, Texas Children's Hospital, Houston, TX, USA** (Prof A M Vogel); **Intensive Care Unit, Bufalini Hospital, AUSL della Romagna, Cesena, Italy** (M V Antonini CCP MSc); **Centre for Human & Applied Physiological Sciences, School of Basic & Medical Biosciences, Faculty of Life Sciences & Medicine, King's College London, London, UK** (N Barrett MBBS); **Department of Critical Care and Nephrology, Guy's and St Thomas' NHS Foundation Trust, London, UK** (N Barrett, Prof M Ostermann MD); **Division of Pediatric Neurology** (G M Benedetti MD), **Division of Pediatric Critical Care Medicine** (S M Gorga MD, R P Barbaro MD), **and Susan B Meister Child Health Evaluation and Research Center, Division of General**

Introduction

Extracorporeal membrane oxygenation (ECMO) support has been delivered to patients with life-threatening illness for more than 50 years.[1] Over time, device technology has evolved, indications for ECMO support have expanded,[2] and ECMO clinical studies and trials in adults[3–5] and children (NCT05405426, NCT05388708, NCT05041712, and NCT06080074) have increased in numbers. Unfortunately, each ECMO clinical study has different adverse event definitions,[3–5] which impedes data assimilation across trials.

When similar definitions are applied to children and adults supported by ECMO, rates of adverse events such as intracranial haemorrhage are consistently higher among children than adults.[3,6,7] This disparity in consequences of care is multifactorial, but one reason is that most ECMO devices are designed and optimised for the adult body size and then adapted to children without specific regulatory clearance.[8] Uniform definitions of adverse events across ECMO registries, studies, and trials are needed to facilitate rigorous safety evaluation of ECMO devices for children.

The Academic Research Consortium (ARC), an international collaborative forum developed to facilitate the creation of stakeholder-driven consensus definitions and nomenclature for clinical trials of medical devices. The ECMO-CENTRAL (Core Elements Needed for Trials Regulation And quality of Life) ARC had a primary aim to develop harmonised definitions of paediatric ECMO adverse events for use in clinical trials and regulatory device evaluation. The secondary aim was to measure the perception of definition appropriateness and feasibility among a diverse group of stakeholders. ECMO-CENTRAL ARC also aimed for international adoption of the resulting consensus-based adverse event definitions through community engagement of stakeholders.

Pediatrics (R P Barbaro), **Department of Pediatrics, University of Michigan, Ann Arbor, MI, USA; Section of Transplantation Surgery, Department of Surgery** (A Rojas-Peña MD), **and Extracorporeal Life Support Lab, Department of Surgery Research** (A Rojas-Peña), **University of Michigan, Ann Arbor, MI, USA; Penn Nursing, University of Pennsylvania, Philadelphia, PA, USA** (A Bettencourt PhD); **Department of Medicine, School of Medicine, John Hopkins University, Baltimore, MD, USA** (Prof D Brodie MD); **TecSalud, University Hospitals School of Medicine, Monterrey Institute of Technology and Higher Education, Monterrey, Mexico** (R Gómez-Gutiérrez MD); **Australian and New Zealand Intensive Care Research Centre, School of Public Health and Preventive Medicine, Monash University, Melbourne, VIC, Australia** (Prof C Hodgson PhD); **Intensive Care Unit, Alfred Hospital, Melbourne, VIC, Australia** (Prof C Hodgson); **Department of Critical Care, University of Melbourne, Parkville, VIC, Australia** (Prof C Hodgson); **Department of Cardiac Anaesthesia, Cardiothoracic and Neuroscience Centre, All India Institute of Medical Sciences, New Delhi, India** (Prof P M Kapoor MD); **Skaggs School of Pharmacy and Pharmaceutical Sciences, University of California, San Diego, CA, USA** (Prof J Le PharmD); **Cardiothoracic Intensive Care Unit, National University Health System, Singapore** (Prof G MacLaren MSc); **Department of Pediatrics, Brooke Army Medical Center, San Antonio, TX, USA** (E R O'Neil MD); **Division of Pediatric Critical Care, Emory University, Atlanta, GA, USA; and Children's Healthcare of Atlanta, Atlanta, GA, USA** (Prof M L Paden MD); **Neonatal Unit, The Royal Hospital for Children, Glasgow, UK** (N Patel MD); **Division of Pediatric Critical Care Medicine, St Louis Children's Hospital** (A S Said MD), **and Institute for Informatics** (A S Said), **Washington University School of Medicine in St Louis, St Louis, MO, USA; Pediatric**

Key messages

- Consensus adverse event definitions for extracorporeal membrane oxygenation (ECMO) in children have not been established, but when similarly defined adverse event classifications were applied in children and adults, it was consistently shown that adverse events were at least two times more common in children
- The ECMO-CENTRAL (Core Elements Needed for Trials Regulation and quality of Life) academic research consortium (ARC) collaborated with international leaders to create consensus definitions for adverse events that were harmonised to existing definitions and gathered formative input and endorsement of research societies and professional organisations
- The resulting adverse event definitions have been disseminated to stakeholders for use in research and regulatory evaluation of devices
- The ECMO-CENTRAL ARC will facilitate child-focused ECMO research and device development to improve the safety of ECMO support for children

Methods

Project design

The ECMO-CENTRAL ARC addressed paediatric patients aged 0 days to 18 years supported by ECMO.[9] This project was done between December, 2022, and July, 2023, using methods informed by previous ECMO consensus projects,[11] ARC processes,[10,12–15] and the EQUATOR Network's Core Outcome Set-Standardised Protocol Items (figure).[16]

The ECMO-CENTRAL ARC leadership (co-chairs PMAA, RPB and members MDN, AMV, AL, and PJDN) developed a list of stakeholders with input from the ARC board, Mechanical Circulatory Support (MCS) ARC chairs, US Food and Drug Administration, and principal investigators of clinical trials in paediatrics, critical care, and ECMO. A diverse group of stakeholders—including trialists, clinicians, patients and families, statisticians, biomedical engineers, device developers, and members of regulatory agencies—were involved in drafting definitions of adverse events that were harmonised with relevant existing ARC definitions, clinical registries, and other definitions adapted for paediatric ECMO adverse events. The ARC leadership met with stakeholders in structured focus groups (appendix p 25). Each meeting included discussion on characterisation of objectives, scope, candidate adverse events, and methods, followed by open-ended questions regarding the process, candidate adverse events, and expert panel members.

Expert panel members were identified based on a combination of the following criteria: academic achievement (competitive grant funding for ECMO research, ECMO clinical trial involvement and publication, paediatric ECMO publication, or publication of definitions of adverse events); direct paediatric ECMO clinical

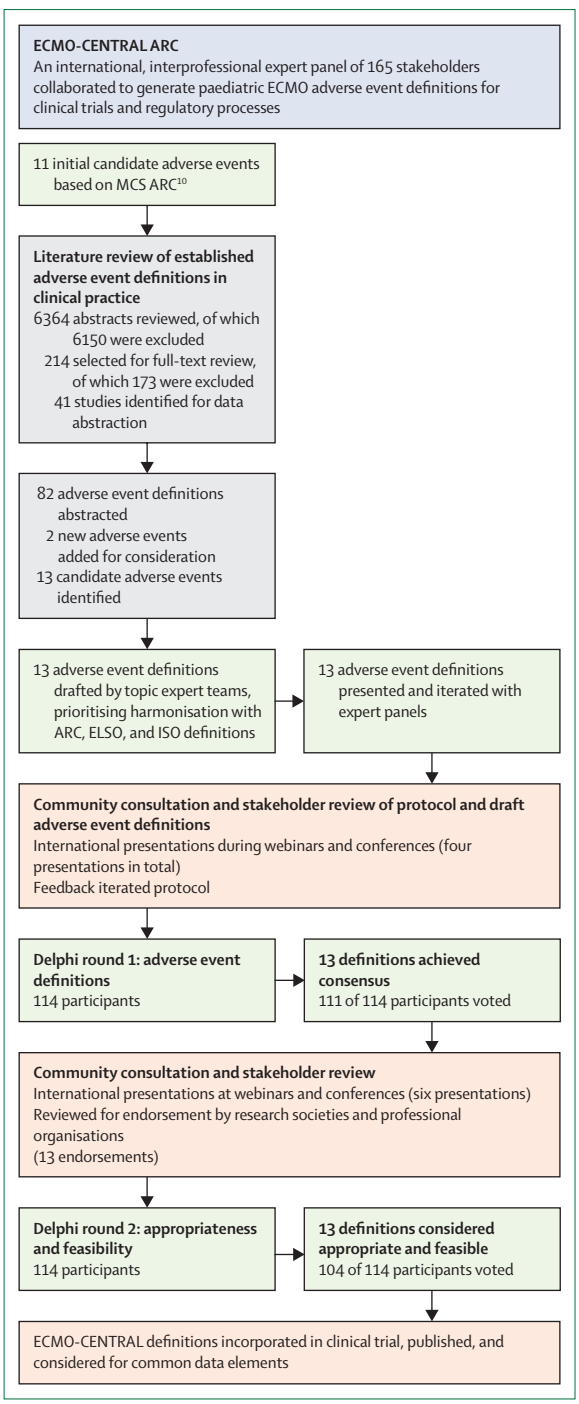

Figure: **Timeline and process map of ECMO-CENTRAL ARC**
The endorsing research societies and professional organisations are shown in the appendix (p 66). ECMO=extracorporeal membrane oxygenation. ECMO-CENTRAL=ECMO Core Elements Needed for Trials Regulation And quality of Life. MCS=mechanical circulatory support. ARC=academic research consortium. ELSO=Extracorporeal Life Support Organization. ISO=Organization for Standardization.

experience; leadership in research societies and professional organisations relevant to ECMO care; and recommendations of selected experts. Attention was given to gender, global representation, individuals from groups historically underrepresented in science and medicine, and diversity in clinical experience that ranged from trainees to established leaders to promote the next generation of researchers in the field (appendix pp 26–32).

Candidate paediatric ECMO adverse events were adapted from the MCS ARC[10] by stakeholders to identify the topics for discussion (appendix p 33). For each candidate adverse event, topic expert teams consisted of 5–15 people with relevant expertise and assigned leaders (GMB, AW, AR-P, SMG, FS, ERO'N, and ASS; appendix pp 34–35). Five investigators (MVA, SK, MLF, JFD, and ERO'N; names and affiliations listed in the appendix pp 3–8) searched PubMed for literature published between Jan 1, 1988, and Feb 20, 2023. The ECMO-CENTRAL ARC search strategy for the adverse events (appendix p 33) was developed with a medical librarian (MLC, name and affiliation listed in the appendix appendix pp 3–8) and was informed by Population, Intervention, Comparator, Outcome, Study design questions that addressed relevance and transferability of the adverse event in paediatric ECMO (appendix p 9). Titles and abstracts were independently reviewed by two investigators and if there was disagreement, a third investigator adjudicated. Full text was reviewed for selected articles. The topic expert leads were asked to supplement the review with established adverse event definitions used in paediatric clinical care and trials.

Topic experts were responsible for developing a draft for each adverse event definition that prioritised harmonisation with previous ARC publications,[10,13–15] the Extracorporeal Life Support Organization (ELSO) Registry, and the International Organization for Standardization (ISO) definitions,[17–19] unless the definitions were considered insufficient for a clinical trial in paediatric ECMO. Adverse event grading was modelled after the standard description for common terminology criteria for adverse events (CTCAE version 5),[20] ranging from grade 1 (mild) to grade 5 (death). Draft definitions for each of the adverse events were first shared with the ECMO-CENTRAL ARC leadership and then presented to the expert panel at adverse event discussion meetings.

Draft adverse event definitions were presented by topic expert leads alongside existing ARC, paediatric ECMO, and other review-identified definitions in March, 2023. A 45-min expert panel discussion was moderated by the co-chairs with input from the panel via open dialogue, direct communication, and an anonymous survey that was available for all sessions. Iterations to the draft definitions were crafted in response to feedback from the expert panel.

The modified Delphi process

All members of the ECMO-CENTRAL expert panel contributed to the development of draft definitions. Additionally, expert panel members were eligible to vote in the consensus process, provided that they did not work for industry or a regulatory agency, were available during the voting window, and provided conflict of interest information details. A-priori expectations were set as a minimum of 50% voting achieved at each round, with 80% or more of voting members in agreement within three rounds of voting. Conflict of interest disclosures were obtained from all voting expert panel members before participating in the modified Delphi process (appendix p 10). If the Delphi process did not meet consensus, the ECMO-CENTRAL ARC leadership group would consider whether a representative adverse event definition could be derived from the MCS ARC or ELSO registry definitions,[15] which would be labelled as representative adverse event definitions.

The ECMO-CENTRAL ARC planned up to three rounds of email-linked, internet-based anonymous surveys conducted as a modified Delphi process. For each round, voters were presented with adverse event definitions and rationale for these definitions. Adverse events were scored by respondents using the RAND/UCLA Appropriateness Method to establish the level of agreement on a scale of 1 (strongly disagree) to 9 (strongly agree), with scores 1–3 representing disagreement, 4–6 equipoise, and 7–9 agreement.[21,22] Comments were requested to suggest edits or to explain scoring. The results and comments were shared with expert panel members between voting rounds. Agreement was defined a priori as 80% or more of the experts rating a recommendation or consensus statement as 7–9.

The definitions were shared with paediatric ECMO researchers and clinicians at international meetings for

	Discussion meeting	Votes in agreement
Neurological	Meeting 1	97/106 (92%)
Bleeding	Meeting 3	94/108 (87%)
Device malfunction	Meeting 3	95/104 (91%)
Acute kidney injury	Meeting 2	96/105 (91%)
Haemolysis	Meeting 1	85/104 (82%)
Infection	Meeting 4	96/108 (89%)
Vascular access-associated injury	Meeting 2	98/107 (92%)
Non-CNS thrombosis	Meeting 4	95/107 (89%)
Hepatic dysfunction	Meeting 4	95/107 (89%)
Right heart failure during venovenous ECMO	Meeting 2	89/104 (86%)
Left ventricular overload	Meeting 4	95/105 (91%)
Inadequate support as lactic acidaemia	Meeting 3	93/107 (87%)
Inadequate support as hypoxaemia	Meeting 3	88/106 (83%)

Data are n (%) or n, where n is the number of expert votes. Meetings 1, 2, 3, and 4 took place on March 1, March 2, March 7, and March 8, respectively. 111 (97%) of 114 expert panel members who were eligible to vote responsed. The percentage agreement includes respondents who voted 7, 8, or 9 on a 9-point Likert scale. The denominator varies because topic experts without expertise in ECMO management were able to select "unable to provide an informed response". ECMO=extracorporeal membrane oxygenation.

Table: **Consensus on adverse events in paediatric ECMO**

Intensive Care Unit, Department of Pediatrics, University Children's Hospital Queen Fabiola–University Hospital of Brussels, Brussels, Belgium (A Willems MD); **Department of Perfusion, University Hospital Gasthuisberg, Leuven, Belgium** (L Vercaemst RN ECCP); **Wallace H Coulter Department of Biomedical Engineering, Georgia Institute of Technology, Atlanta, GA, USA** (Prof A P Yoganathan PhD); **The Heart Institute, Cincinnati Children's Hospital Medical Center, University of Cincinnati, Cincinnati, OH, USA** (Prof A Lorts MD)

Correspondence to: Ryan P Barbaro, Division of Pediatric Critical Care Medicine, University of Michigan, Ann Arbor, MI 48109, USA **barbaror@med.umich.edu**

See **Online** for appendix

For the **Extracorporeal Life Support Organization Registry Data Definitions** see https://www.elso.org/Registry/DataDefinitions,Forms,Instructions.aspx

Panel 1: **Neurologic adverse events**

Neurologic adverse events are identified during extracorporeal membrane oxygenation (ECMO) support or during the first post-ECMO intracranial imaging or autopsy (excluding pre-existing strokes, intracranial haemorrhage, and diffuse ischaemic injury; neurological adverse events are particularly relevant to neonates in whom extra-axial haemorrhage might be a complication of a difficult or traumatic delivery). Reporting should include the number of hours of support after which the adverse event occurred.

Neurological adverse events

A Ischaemic stroke: evidence of new focal neurological deficits corresponding to a vascular territory or territories within the brain meeting one major criterion or two minor criteria:
- Major ischaemic stroke: radiographical (cranial ultrasound, CT, or MRI) or pathological (autopsy) confirmation of ischaemic stroke
- Minor ischaemic stroke: focal neurological signs without radiographical or pathological confirmation, and with two or more of the following:
  - Persistent focal neurological dysfunction localising to vascular territories (can include new focal electroencephalography changes)
  - Persistent focal neurological dysfunction temporally correlated with known clot dislodgement
  - Diagnosis of stroke by expert vascular neurologist or paediatric neurocritical care specialist

B Intra-axial haemorrhage without mass effect: radiographical (cranial ultrasound, CT, or MRI) or pathological (autopsy) confirmation; includes petechial bleeding into the brain parenchyma, microhaemorrhage, and haemorrhagic conversion of ischaemic stroke without space-occupying effect (not caused by trauma); includes low-grade intraventricular haemorrhage grade 1–2

C Intra-axial haemorrhage with mass effect: radiographical (cranial ultrasound, CT, or MRI) or pathological (autopsy) confirmation; includes parenchymal haemorrhage and intraventricular haemorrhage grade 3–4 resulting in mass effect or obstructive hydrocephalus (not caused by trauma)

D Extra-axial haemorrhage: bleeding into the subarachnoid, subdural, or epidural space that is confined within defined compartment (not caused by trauma)

E Mixed intra-axial and extra-axial haemorrhage: radiographical (cranial ultrasound, CT, or MRI) or pathological (autopsy) confirmation; includes components of extra-axial haemorrhage and either intra-axial haemorrhage with or without mass effect (not caused by trauma)

F Diffuse ischaemic injury: radiographically confirmed by cranial ultrasound, CT, or MRI or pathologically by autopsy; acute diffuse brain injury; patterns of injury can include global, cortical, subcortical, and watershed distribution

G Seizure: evidence of seizure not attributable to pre-existing epilepsy, meeting one major criterion or two minor criteria:
- Major seizure: electroencephalographic (EEG) seizure
- Minor seizure: a clinical event of transient signs or symptoms with a clear start and finish, presumed to be due to abnormal excessive or synchronous neuronal activity in the brain; if this event occurred when EEG was not in place, then confirmation is required with two of the following:
  - EEG with evidence of focal neuronal dysfunction and cortical hyperexcitability (eg, recurrent regional epileptiform abnormalities)
  - Diagnosed by an expert neurologist based on semiology or response to treatment

H Status epilepticus: more than 5 min of continuous convulsive activity (clinical or EEG), more than 10 min of continuous non-convulsive seizure (clinical or EEG), or more than 12 min in total in a 60 min epoch (EEG)

J Neuropathy: new neuropathy or plexopathy attributable to ECMO; includes femoral, sciatic, or phrenic neuropathy, brachial or lumbosacral plexopathy, or second or third order Horner syndrome

K Death by neurological criteria: meeting criteria per institutional policy or applicable legal statute

Patient consequence

Patient consequences are organised as types according to the change in Pediatric Cerebral Performance Category (PCPC) with six, scores 1–6[25] from baseline (pre-morbid or pre-ECMO and can be obtained retrospectively) to hospital discharge. A PCPC score of 4 is a patient with severe disability who is conscious but dependent on others for daily support because of impaired brain function. If a patient qualifies for multiple types, select the most severe. For example, if a patient has a 1-point increase from a baseline PCPC score of 4 to a PCPC score of 5 at hospital discharge, classify as a type 3 consequence rather than type 2. If a patient's pre-ECMO PCPC score is 5 and their outcome is death, classify as type 4 rather than type 2 (increase of 1 point) or type 3 (PCPC score >4):
- Type 1: no change in PCPC score
- Type 2: 1-point increase in PCPC score
- Type 3: 2-point increase in PCPC or an increase to PCPC score higher than 4
- Type 4: fatal

iterative feedback preceding and between voting rounds. After formative insights from leaders of research societies and professional organisations, a draft manuscript was submitted to those organisations with a stake in paediatric ECMO research and device development for their edits and endorsement. To gain a perception of definition appropriateness and feasibility, selected and adapted items[23] were applied from the validated Intervention Appropriateness Measure and Feasibility of Intervention Measure

Panel 2: **Bleeding adverse event**

Bleeding is adjudicated at least daily during extracorporeal membrane oxygenation (ECMO) support and for up to 6 h after decannulation

Bleeding adverse event types

Type 1: overt bleeding associated with any one of the following:
- Measurable blood loss of less than 2 mL/kg/h (up to 20 mL/kg in any 24 h period; for patients >50 kg, total measurable blood loss of <100 mL/h for 2 h or more, up to 1000 mL in any 24 h period) or bloody dressings changed no more frequently than every 6 h
- Clinically insignificant bleeding (eg, from the endotracheal tube, nasogastric tube, in-dwelling urinary catheter, mucosal, or cutaneous bleeding including petechiae or purpura)
- Actions included observation, except for the use of local haemostatic agents only (no administration of systemic pharmacological agents or blood products)

Type 2: overt bleeding associated with any one of the following:
- Measurable blood loss of 2–5 mL/kg/h for 2 h or more (up to 40 mL/kg in any 24 h period; for patients >50 kg, total measurable blood loss of 100–250 mL/hr for 2 h or more, up to 2000 mL in any 24 h period)
- Haemoglobin drop: a drop of 1–3 g/dL (10–30 g/L) in less than 12 h, but with no impaired oxygen-delivery capacity: lactate less than 3 mmol/L (<27 mg/dL) or falling over 12 h
- Gastrointestinal bleeding, pulmonary haemorrhage, mediastinal cannulation site bleeding, or surgical site bleeding more than type 1, but does not meet the criteria of type 3 bleeding

Type 3: overt bleeding associated with any two of the following:
- More than 5 mL/kg/h for 2 h or more (or >40 mL/kg in any 24 h period; for patients >50 kg, total measurable blood loss of >250 mL/h for 2 h or more, or >2000 mL in any 24 h period)
- Haemodynamic instability associated with the bleeding in the absence of other causes
  - Increase in vasoactive agents in response to haemodynamic instability
  - Volume administration to maintain pump preload
- Severe haemoglobin drop: a drop (absolute value) of more than 3 g/dL (>30 g/L) within 12 h, with impaired oxygen-delivery capacity evidenced by lactate of more than 3 mmol/L (>27 mg/dL) and increasing in two consecutive measurements at least 2 h apart
- Bleeding causing cardiac tamponade or compartment syndrome (abdomen or muscular compartment)

Type 4: fatal

instruments[24] for the adverse event definitions (appendix p 36).

Results

The literature search identified 6364 manuscripts, of which 214 (3%) were selected for full-text review. 82 definitions extracted from 41 articles informed the development of definitions for adverse events (appendix pp 33, 37–48). The informing references were available to topic expert groups during development of draft definitions.

ECMO-CENTRAL ARC engaged 165 stakeholders and expert panel members from 94 institutions and 23 countries, and 114 individuls were eligible to vote. 54 (47%) of 114 voting members were women and 60 (53%) were men, 24 (21%) self-identified as belonging to groups historically underrepresented in science and medicine, and 36 (32%) were academic professors. In terms of primary professional specialty, voting members included 17 (15%) paediatric cardiac intensivists or cardiologists, eight (7%) congenital heart surgeons, two (2%) paediatric surgeons, nine (8%) neonatologists, 32 (28%) paediatric intensivists, nine (8%) ECMO specialists (ie, registered nurse, respiratory therapist, or perfusionist with additional relevant training), 19 (17%) adult-focused specialists, and 18 (16%) members from other specialties.

Initially proposed adverse events derived from the MCS ARC (appendix p 33) were refined based on expert panel input for the 13 adverse event definitions listed in the table. Four structured adverse event definition discussion meetings lasting 3 h each were held by video conference between March 1 and March 8, 2023. Topics for discussion were announced in advance, and each topic expert group presented the draft definitions at one or more of the meetings. Although consensus voting was anonymous, feedback on the draft definitions was obtained by open dialogue, direct communication, and an anonymous survey at each meeting. Iterations to the definitions were made by the topic leads and co-chairs, and draft definitions were presented a second time as required. Consensus voting took place from April 10 to April 24, 2023. At the first round of voting, 111 (97%) of 114 participants voted, and consensus of more than 80% agreement was achieved for each definition (table).

The ECMO-CENTRAL ARC held community feedback meetings with ELSO, the International Chapters of ELSO, and PediECMO (a research collaborative among the Pediatric Acute Lung Injury and Sepsis Investigators and ELSO) from March 17 to June 13, 2023. Edits from professional and research organisations were incorporated into the definitions, and endorsement was (endorsing professional societies and research networks are listed in the appendix p 66). The second round of Delphi survey

For the **ECMO-CENTRAL ARC meeting recordings** see ecmo-central.medicine.umich.edu

Panel 3: **Mechanical adverse events**

Device malfunction adverse events occur during ongoing extracorporeal membrane oxygenation (ECMO) support only. Reporting should include the number of hours of ECMO support at which the adverse event occurred.

Device component and replacement

ECMO device components that can cause malfunction include:

- A Blood pump
- B Membrane lung
- C Cannula (cannula dysfunction does not require replacement, but requires demonstration of visual evidence of thrombus within the cannula, leading to an intervention for thrombi removal)
- D Tubing (tubing dysfunction can include replacement, segment exchange, or cutout)
- E Tubing connectors
- F Heater unit
- G ECMO console

In some instances, rather than substituting the defective component, the whole ECMO system will need replacing due to the defective component. In other instances, the whole system will need replacing due to a dysfunction that cannot be localised to a specific component.

Device malfunction type, according to degree of failure, relatedness, and abruptness

- Type 1—structural failure: anything that affects the integrity of the device separation of blood and air, leading to blood leak, air entrainment, or embolus, and managed with replacement or separation from ECMO or resulting in death
- Type 2—functional failure: failure or presumed failure of normal mechanical operation and managed with replacement or separation from ECMO, or resulting in death; failure or presumed failure of normal operation of the blood pump, some membrane lungs, heat exchanger, ECMO console, or the whole system (when integrated)
- Type 3—device dysfunction: failure of the device to perform its dedicated task at optimal parameters, managed with replacement or separation from ECMO, or resulting in death
- Type 4—device replacement: replacement of a device without documented failure, often due to a centre-specific protocol regarding longevity of use

Patient consequences

- Grade 1: transient, asymptomatic, or mild symptoms; clinical or diagnostic observations only
- Grade 2: maintenance of patient vital signs that require an escalation in non-extracorporeal life support
- Grade 3: hypotension or hypoxaemia despite medical interventions
- Grade 4: cardiopulmonary arrest requiring cardiopulmonary resuscitation
- Grade 5: fatal

(May 8–23, 2023) assessed the appropriateness and feasibility of ECMO-CENTRAL ARC definitions of adverse events. Among 114 eligible voters, 104 (91%) voted. 84–96% of respondents deemed ECMO-CENTRAL ARC adverse events to be appropriate, and 82–96% regarded the adverse events to be feasible to measure (appendix p 67).

Consensus definitions

The expert panel defined 13 adverse event definitions: neurological, bleeding, device malfunction, acute kidney injury, haemolysis, infection, vascular access-associated injury, non-CNS thrombosis, hepatic dysfunction, right heart failure during venovenous ECMO, left ventricular overload, lactic acidaemia, and hypoxaemia. Definitional structure varied across adverse events.

The neurological adverse event definition has two components (panel 1). The first component defines the types of adverse events, including intracranial bleeding, stroke, seizure, and peripheral nerve dysfunction, whereas the second component characterises changes in patient functioning. The bleeding definition includes four levels of worsening patient consequence (panel 2). Mechanical adverse events characterise the component devices involved, the type of device malfunction, and the patient consequences independently (panel 3). Adverse event definitions for acute kidney injury, haemolysis, infection, vascular access-associated injury, non-CNS thrombosis, hepatic dysfunction, right heart failure during venovenous ECMO, left ventricular overload, lactic acidaemia, and hypoxaemia can be found in the appendix (pp 49–58). The adverse events are also presented with grading criteria modelled on CTCAE for clinical trials (appendix pp 59–66).

Discussion

The ECMO-CENTRAL ARC generated adverse event definitions harmonised to those established for clinical trials and regulatory evaluation of paediatric ECMO-related devices. To create the ECMO-CENTRAL ARC adverse event definitions, experts reviewed the existing literature, topic experts drafted the definitions, and stakeholders (including trialists, clinicians, patients and families, statisticians, biomedical engineers, device developers, and members of regulatory agencies) provided revision input. These definitions were harmonised with adverse event definitions and further refined with community input. The ECMO-CENTRAL ARC definitions and associated rationale are included in the appendix (pp 11–24, 49–58). These rigorously and holistically crafted definitions reached consensus among the voting expert panel of 114 members from across the world and were assessed to be appropriate and feasible to record the targeted adverse events.

Over the past 50 years, ECMO has been delivered to more than 80 000 children as a recommended life-support strategy for refractory cardiopulmonary failure.[6,26–28] ECMO remains the sum of a myriad assembled component technologies that are cleared for use in adults and adapted to children, which might contribute to the disparity in adverse events sustained by ECMO-supported children compared with adults. Overcoming this disparity will require organised collaboration, infrastructure, and innovation that is emerging in the ECMO speacialty and borrows from the successful ARC model.[9–11] The ARC was designed to serve as a stakeholder-driven platform for generating, harmonising, and disseminating consensus definitions and standard nomenclature for pivotal clinical trials of medical devices.[29] With comprehensive cataloguing of adverse events associated with device-related interventions,[29] the ARC has facilitated documentation of safety profiles for emerging devices designed for adults. The ECMO-CENTRAL ARC produced a broad list of ECMO-associated adverse events, leveraging existing definitions appropriate for critically ill children by advancing a reporting standard for safety documentation of existing and emerging devices.

Organising frameworks for ECMO clinical research began with standardised nomenclature in 2017,[9] followed by core outcomes for adults,[11] which include adverse events of intracranial haemorrhage, bleeding, mortality, and measures of life effects. The ECMO-CENTRAL ARC adds a comprehensive list of adverse events and uniquely focuses on children. There was some anticipated overlap of core outcomes and adverse events; in these cases, harmonisation was prioritised. For example, intracranial haemorrhage was defined by ECMO core outcomes,[11] the MCS ARC,[10] the ECMO-CENTRAL ARC, and the Society of Thoracic Surgeons Interagency Registry for Mechanically Assisted Circulatory Support. This coordination eliminates unnecessary differences in definitions and facilitates consistency in reporting for both children and adults when possible. However, harmonisation was not always achievable. While the MCS ARC definitions were the foundation for the ECMO-CENTRAL ARC, when incongruencies between the technologies and the population existed, the definitions were adapted. For example, MCS ARC defined device failure in the context of integrated mechanical circulatory support systems, a construct that did not translate to the component technologies that are assembled for ECMO.

Importantly, adverse events such as acute kidney injury and bleeding have been defined for complications that occur more broadly across critical illness.[30,31] The established Kidney Disease Improving Global Outcomes (KDIGO) definition was integrated into the ECMO-CENTRAL ARC with the guidance of KDIGO expert contributors.[30] In the case of bleeding, there were multiple iterative definitions including MCS,[15] the Bleeding Assessment Scale in Critically Ill Children,[31] trauma, and with those used in an ongoing clinical trial of indication-based red blood cell transfusion in ECMO (NCT05405426).[32,33] Authors of each of the definitions contributed to the ECMO-CENTRAL ARC to harmonise existing definitions, while also ensuring appropriateness of the comprehensive collection of adverse event definitions to children receiving ECMO support.

In addition to the assessments of appropriateness and feasibility, which were undertaken as a part of the ECMO-CENTRAL ARC process, future work is required to establish the validity and reproducibility of data abstracted as meeting these definitions. Epidemiology of the ECMO-CENTRAL ARC adverse events in children who receive ECMO will be established in observational studies and captured in clinical trials.

Future steps for the ECMO-CENTRAL ARC include applying for inclusion in the common data elements of the US National Institutes of Health (NIH) and UK National Institute for Health and Care Research and incorporation of definitions into the ELSO Registry to encourage dissemination. The ECMO-CENTRAL ARC will also work to include definitions into standard informatics vocabularies, such as Logical Observation Identifiers Names and Codes or Systematized Nomenclature of Medicine—Clinical Terms. Such digitisation and capture of adverse events associated with ECMO from electronic health records could serve as digital quality measures to monitor the delivery of ECMO-related care across sites. Future investigative efforts should assess the feasibility, accuracy, and reproducibility of assigning relatedness to adverse events of ECMO.[34] The ECMO-CENTRAL ARC expert panel anticipated that clinician adjudication would be required to attribute the relatedness of adverse events to ECMO devices, patient illness, or clinician management. ECMO-CENTRAL ARC definitions will be evaluated for updating every 2–5 years according to the ARC guidelines of updates.

The ECMO-CENTRAL ARC has paired rigorous development with methodical stakeholder involvement, dissemination, and endorsement by prominent research societies and professional organisations. These efforts were undertaken because ECMO-supported children have a more than two-fold risk of experiencing adverse events compared with ECMO-supported adults. This disparity was a call to action for the expert panel. Increased device development, scientific investigation, and funding have positioned paediatric ECMO at the precipice of an era of improved outcomes, but to capitalise on these investments the field needs to be organised. The application of ECMO-CENTRAL ARC definitixons across clinical research for ECMO-supported children will maximise return on investment as a standard for data capture and assimilation.

Contributors

PMAA and RPB conceived the study. PMAA, MDN, AC, AMV, AL, PJdN, and RPB designed the study. PMAA, MDN, AC, GMB, SMG, ERO'N, AR-P, ASS, FS, AW, and RPB drafted the initial manuscript. All authors analysed and interpreted the results, critically edited the manuscript, approved the final work, and are accountable for the accuracy and integrity of the work.

Declaration of interests
PMAA reports institution funding from the US Department of Defense (PRMRP clinical trial award #W81XWH2210301), National Institutes of Health (NIH R13HD104432), and the US Food and Drug Administration (FDA Grant Number U01FD004979/U01FD005978); and is the Treasurer for Extracorporeal Life Support Organization (ELSO). MDN is a member of the advisory board for Eurosets and a member of the European ELSO Chapter Steering Committee. MVA, NB, and LV are members of the European ELSO Chapter Steering Committee. DB received research support and consulting fees from LivaNova; has been on medical advisory boards for Xenios, Medtronic, Inspira, and Cellenkos; is the President-Elect of ELSO and the Chair of the Executive Committee of the International ECMO Network (ECMONet); and writes for *UpToDate*. RG-G is a member of the Latin American ELSO Chapter Steering Committee. PMK is a member or the Southwest Asia and Africa ELSO Chapter Steering Committee. GM is President of ELSO. MLP is the Past President of ELSO. AR-P is a consultant for Terumo CVS; and his institution receives funding from the NIH (R01 HL141509 and R01 HL155100). AL is a consultant for Bayer, Abbott, Abiomed, Berlin Heart, and Syncardia; is the primary investigator for the Berlin ACTIVE driver trial; and is an Executive of the ACTION Network. RPB reports a grant from the NIH (R01 HL153519) and is a member of the ELSO Board of Directors and Co-Chair of PediECMO.

Acknowledgments
We thank all our collaborators who are listed in the appendix (pp 3–8).

References

1 Hill JD, O'Brien TG, Murray JJ, et al. Prolonged extracorporeal oxygenation for acute post-traumatic respiratory failure (shock-lung syndrome). Use of the Bramson membrane lung. *N Engl J Med* 1972; **286:** 629–34.
2 Pollack BE, Kirsch R, Chapman R, Hyslop R, MacLaren G, Barbaro RP. Extracorporeal membrane oxygenation then and now; broadening indications and availability. *Crit Care Clin* 2023; **39:** 255–75.
3 Combes A, Hajage D, Capellier G, et al. Extracorporeal membrane oxygenation for severe acute respiratory distress syndrome. *N Engl J Med* 2018; **378:** 1965–75.
4 Belohlavek J, Smalcova J, Rob D, et al. Effect of intra-arrest transport, extracorporeal cardiopulmonary resuscitation, and immediate invasive assessment and treatment on functional neurologic outcome in refractory out-of-hospital cardiac arrest: a randomized clinical trial. *JAMA* 2022; **327:** 737–47.
5 Thiele H, Zeymer U, Akin I, et al. Extracorporeal life support in infarct-related cardiogenic shock. *N Engl J Med* 2023; **389:** 1286–97.
6 Tonna JE, Boonstra PS, MacLaren G, et al. Extracorporeal life support organization registry international report 2022: 100,000 survivors. *ASAIO J* 2024; **70:** 131–43.
7 Dalton HJ, Reeder R, Garcia-Filion P, et al. Factors associated with bleeding and thrombosis in children receiving extracorporeal membrane oxygenation. *Am J Respir Crit Care Med* 2017; **196:** 762–71.
8 US Food and Drug Administration. Premarket approval of pediatric uses of devices FY 2019–FY 2020. 2020. https://www.fda.gov/media/171143/download (accessed June 10, 2023).
9 Conrad SA, Broman LM, Taccone FS, et al. The extracorporeal life support organization Maastricht treaty for nomenclature in extracorporeal life support. A position paper of the extracorporeal life support organization. *Am J Respir Crit Care Med* 2018; **198:** 447–51.
10 Kormos RL, Antonides CFJ, Goldstein DJ, et al. Updated definitions of adverse events for trials and registries of mechanical circulatory support: a consensus statement of the mechanical circulatory support academic research consortium. *J Heart Lung Transplant* 2020; **39:** 735–50.
11 Hodgson CL, Fulcher B, Mariajoseph FP, et al. A core outcome set for research in patients on extracorporeal membrane oxygenation. *Crit Care Med* 2021; **49:** e1252–54.
12 Krucoff MW, Mehran R, van Es GA, Boam AB, Cutlip DE. The academic research consortium governance charter. *JACC Cardiovasc Interv* 2011; **4:** 595–96.
13 Mehran R, Rao SV, Bhatt DL, et al. Standardized bleeding definitions for cardiovascular clinical trials: a consensus report from the Bleeding Academic Research Consortium. *Circulation* 2011; **123:** 2736–47.
14 Généreux P, Piazza N, Alu MC, et al. Valve academic research consortium 3: updated endpoint definitions for aortic valve clinical research. *J Am Coll Cardiol* 2021; **77:** 2717–46.
15 Lansky AJ, Messé SR, Brickman AM, et al. Proposed standardized neurological endpoints for cardiovascular clinical trials: an Academic Research Consortium initiative. *J Am Coll Cardiol* 2017; **69:** 679–91.
16 Kirkham JJ, Gorst S, Altman DG, et al. Core Outcome Set-STAndardised Protocol Items: the COS-STAP statement. *Trials* 2019; **20:** 116.
17 International Organization for Standardization. Medical devices—quality management systems—requirements for regulatory purposes, 3rd edn. 2016. https://www.iso.org/obp/ui/en/#iso:std:iso:13485:ed-3:v1:en (accessed Jan 15, 2023).
18 International Organization for Standardization. Biological evaluation of medical devices—part 11: tests for systemic toxicity, 3rd edn. 2016. https://www.iso.org/obp/ui/#iso:std:iso:10993:-11:ed-3:v1:en (accessed Jan 15, 2023).
19 International Organization for Standardization. Cardiovascular implants and artificial organs—blood-gas exchangers (oxygenators)—amendment 1: connectors. 2016. https://www.iso.org/obp/ui/es/#iso:std:iso:7199:ed-3:v1:amd:1:v1:en (accessed Jan 15, 2023).
20 National Cancer Institute. Common terminology criteria for adverse events. Nov 27, 2017. https://ctep.cancer.gov/protocoldevelopment/electronic_applications/docs/ctcae_v5_quick_reference_5x7.pdf (accessed March 9, 2023).
21 Fitch K, Bernstein SJ, Aguilar MD, et al. The RAND/UCLA appropriateness method user's manual. Santa Monica, CA: RAND Corporation, 2001.
22 Diamond IR, Grant RC, Feldman BM, et al. Defining consensus: a systematic review recommends methodologic criteria for reporting of Delphi studies. *J Clin Epidemiol* 2014; **67:** 401–09.
23 Schondelmeyer AC, Bettencourt AP, Xiao R, et al. Evaluation of an educational outreach and audit and feedback program to reduce continuous pulse oximetry use in hospitalized infants with stable bronchiolitis: a nonrandomized clinical trial. *JAMA Netw Open* 2021; **4:** e2122826.
24 Weiner BJ, Lewis CC, Stanick C, et al. Psychometric assessment of three newly developed implementation outcome measures. *Implement Sci* 2017; **12:** 108.
25 Fiser DH, Tilford JM, Roberson PK. Relationship of illness severity and length of stay to functional outcomes in the pediatric intensive care unit: a multi-institutional study. *Crit Care Med* 2000; **28:** 1173–79.
26 Marino BS, Tabbutt S, MacLaren G, et al. Cardiopulmonary resuscitation in infants and children with cardiac disease: a scientific statement from the American Heart Association. *Circulation* 2018; **137:** e691–782.
27 Weiss SL, Peters MJ, Alhazzani W, et al. Executive summary: surviving sepsis campaign international guidelines for the management of septic shock and sepsis-associated organ dysfunction in children. *Pediatr Crit Care Med* 2020; **21:** 186–95.
28 Lorusso R, Whitman G, Milojevic M, et al. 2020 EACTS/ELSO/STS/AATS expert consensus on post-cardiotomy extracorporeal life support in adult patients. *J Thorac Cardiovasc Surg* 2021; **161:** 1287–331.
29 Cutlip DE, Windecker S, Mehran R, et al. Clinical end points in coronary stent trials: a case for standardized definitions. *Circulation* 2007; **115:** 2344–51.
30 Khwaja A. KDIGO clinical practice guidelines for acute kidney injury. *Nephron Clin Prac* 2012; **120:** c179–84.
31 Nellis ME, Tucci M, Lacroix J, et al. Bleeding Assessment Scale in Critically Ill Children (BASIC): physician-driven diagnostic criteria for bleeding severity. *Crit Care Med* 2019; **47:** 1766–72.
32 Nellis ME, Dalton H, Karam O. Quantifiable bleeding in children supported by extracorporeal membrane oxygenation and outcome. *Crit Care Med* 2019; **47:** e886–92.
33 Levy JH, Faraoni D, Almond CS, et al. Consensus statement: hemostasis trial outcomes in cardiac surgery and mechanical support. *Ann Thorac Surg* 2022; **113:** 1026–35.
34 Cutlip DE, Mehran R, Spitzer E, et al. Device and procedure relatedness: viewpoint from members of the ARC steering group. *JACC Cardiovascular Interventions* 2022; **15:** 783–88.

Copyright © 2024 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.